

B.Tech III Year II Semester (R20) Regular & Supplementary Examinations July 2024

SOFTWARE TESTING

(Computer Science and Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) What is meant by testing? Why is it needed? | 2M |
| (b) What are the applications of Path Testing? | 2M |
| (c) Distinguish between Transaction flow and control flow. | 2M |
| (d) Write about any two application of data flow testing. | 2M |
| (e) What is domain span? | 2M |
| (f) Differentiate between boundary point and extreme point. | 2M |
| (g) Write the limitations of path testing. | 2M |
| (h) Illustrate decision tables with examples. | 2M |
| (i) Differentiate between dead state and unreachable state. | 2M |
| (j) How can the graph be represented in matrix form? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 State and explain various dichotomies in software testing. 10M
- OR**
- 3 (a) List the elements of flow graph and explain each element with suitable diagram. 5M
(b) Demonstrate various kinds of loops with respect to path testing using neat diagrams. 5M
- 4 (a) Differentiate various transaction flow testing techniques. 5M
(b) Discuss about complication in transaction-flow testing. 5M
- OR**
- 5 Explain in detail about data flow testing strategies. 10M
- 6 (a) Explain about testing one-dimensional domains. 5M
(b) State and explain various restrictions at domain testing process. 5M
- OR**
- 7 Discuss the domain and interface testing in detail. 10M
- 8 Explain Regular Expression and Flow Anomaly detection. 10M
- OR**
- 9 (a) Write rules of Boolean algebra. 5M
(b) Write the procedure for specification validation. 5M
- 10 (a) Categorize various testing tools necessary for testing. 5M
(b) Explain state testing in detail. 5M
- OR**
- 11 (a) Explain about node reduction algorithm. 5M
(b) Explain good state graph with suitable example. 5M
